

IMPERIAL BEACH NOLF, CA, USA

WMO: 722909

Lat: 32.567N

Lon: 117.117W

Elev: 7

StdP: 101.24

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 93115

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.7	6.7	-9.6	1.7	17.1	-6.4	2.2	15.2	10.9	14.7	9.3	15.1	2.7	90	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	29.9	18.1	27.4	18.7	25.9	18.7	22.1	26.6	21.4	25.4	20.7	24.4	3.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.2	24.3	19.8	14.5	24.0	19.1	13.9	23.4	64.7	26.7	62.2	25.4	59.8	24.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.4	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	13.9	13.6	14.4	15.5	16.8	18.6	20.5	21.0	20.5	18.5	16.1	13.4	(6)
(7)		DBStd	3.56	2.29	2.50	2.25	1.98	1.92	1.82	1.74	2.50	3.04	2.56	2.34	2.18	(7)
(8)		HDD10.0	3	1	1	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	831	138	134	126	90	56	17	3	3	9	28	74	153	(9)
(10)		CDD10.0	2528	123	102	135	163	210	258	324	342	316	264	184	107	(10)
(11)		CDD18.3	314	1	2	3	3	8	24	69	87	74	34	8	1	(11)
(12)		CDH23.3	1317	25	34	37	40	57	38	106	301	344	219	106	10	(12)
(13)	CDH26.7	317	2	8	11	10	19	4	9	58	109	53	33	1	(13)	
(14)	Wind	WSAvg	3.0	2.9	3.1	3.0	3.2	3.3	3.1	3.0	2.9	2.7	2.6	2.8	3.0	(14)
(15)	Precipitation	PrecAvg	191	35	23	38	15	4	2	0	3	6	36	29	(15)	
(16)		PrecMax	423	147	78	138	82	29	13	2	2	38	20	163	112	(16)
(17)		PrecMin	120	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	80	34	22	32	20	6	4	1	0	8	6	36	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	27.7	27.5	28.0	29.7	26.8	27.4	30.7	35.1	31.3	31.0	24.8	(19)
(20)			MCWB	12.1	13.2	14.4	14.3	15.6	20.0	21.1	21.8	22.2	17.4	14.2	10.8	(20)
(21)		2%	DB	23.2	23.3	23.0	23.6	24.1	24.0	25.6	27.5	29.2	27.9	25.9	21.7	(21)
(22)			MCWB	11.6	12.5	13.8	14.3	15.7	18.3	20.6	21.2	20.4	17.2	13.8	12.0	(22)
(23)		5%	DB	21.0	20.5	20.2	21.2	21.2	22.7	24.4	25.9	26.4	25.6	23.4	19.5	(23)
(24)			MCWB	11.8	13.1	14.2	14.8	16.0	17.8	19.8	20.5	21.1	18.1	13.6	12.6	(24)
(25)		10%	DB	18.9	18.6	18.7	19.7	20.0	21.7	23.5	24.8	24.9	23.8	21.2	18.2	(25)
(26)			MCWB	12.5	13.3	14.0	14.8	15.5	17.3	19.2	20.1	20.5	18.0	14.2	12.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.7	17.1	17.6	18.6	20.2	22.2	23.0	23.4	22.1	18.0	16.5	(27)
(28)			MCDWB	18.3	19.4	20.7	21.8	22.9	25.6	26.7	28.4	30.4	26.3	21.5	18.7	(28)
(29)		2%	WB	15.3	15.5	16.1	16.3	17.3	19.1	21.1	21.7	22.4	20.7	16.9	15.6	(29)
(30)			MCDWB	18.0	18.9	20.2	20.3	21.1	23.3	24.7	26.1	27.0	24.8	20.5	18.0	(30)
(31)		5%	WB	14.7	14.7	15.3	15.7	16.4	18.4	20.4	21.0	21.6	19.6	16.3	14.8	(31)
(32)			MCDWB	17.4	18.3	18.6	19.7	20.1	22.0	23.7	25.1	25.7	24.1	20.2	17.5	(32)
(33)		10%	WB	14.0	14.0	14.6	15.2	15.7	17.8	19.8	20.4	20.9	18.6	15.6	13.9	(33)
(34)			MCDWB	16.9	17.5	17.7	19.1	19.2	21.2	22.8	24.3	24.4	22.6	19.6	16.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	9.4	8.3	8.2	6.4	5.2	5.2	5.9	7.5	9.3	10.4	9.9	(35)
(36)			MCDBR	14.3	14.2	11.7	12.7	9.1	6.6	6.4	7.3	9.6	13.1	15.1	12.8	(36)
(37)			MCWBR	7.6	7.9	6.8	7.3	4.9	3.6	3.3	3.5	4.5	7.4	8.5	8.0	(37)
(38)		5% WB	MCDWB	8.9	10.9	9.1	9.9	7.8	6.5	6.5	7.1	8.3	10.4	10.7	9.0	(38)
(39)			MCWBR	6.1	7.0	5.8	6.0	4.4	3.7	3.3	3.6	4.1	6.5	7.3	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.327	0.339	0.352	0.372	0.373	0.400	0.399	0.366	0.368	0.324	0.326	(40)	
(41)		taud	2.556	2.537	2.486	2.453	2.417	2.417	2.424	2.406	2.477	2.448	2.557	2.552	(41)	
(42)		Ebn at Noon	910	932	941	937	915	909	882	880	901	871	891	876	(42)	
(43)		Edh at Noon	82	92	104	112	118	117	116	116	104	99	81	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.33	4.19	5.47	6.59	6.86	7.10	6.94	6.72	5.84	4.59	3.60	3.01	(44)	
(45)		RadStd	0.20	0.34	0.31	0.36	0.53	0.45	0.49	0.32	0.34	0.34	0.24	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (35 neighbors)	+0.65	N/A	-1.57	+0.78	N/A	N/A	N/A	-105	+230	+122	(47)

Nomenclature: See separate page